

MD. RESHAD AL MUTTAKI

EEE GRADUATE | RESEARCH ASSISTANT

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SUMMARY

EEE graduate and Research Assistant focusing on self-powered biosensors, embedded systems, and smart grids, with strong experience in mentoring, leadership, digital logic design, machine learning, and multiple Q1 publications in energy and cybersecurity research.

PROFESSIONAL EXPERIENCE

Research Assistant (RA) - Full Time, BRAC University

Mar 2026 - Present

- Ongoing research on “Advancing Sustainable Biosensing: A Self-powered Biosensor for Real-time and Wireless Health Monitoring”, a fully funded project by BRAC University.
- Developing an e-skin-type biosensor both in simulation and hardware environments.

Student Tutor (ST), BSRM SoE, Dept. of EEE, BRAC University

Jan 2024 - Oct 2025

- Managed and mentored undergraduate students through tutorial and consultation sessions across multiple courses, fostering strong leadership, mentorship, and instructional skills.

Student Mentor, FYAT, OAA, BRAC University

Jan 2024 - Jan 2025

- Provided one-on-one mentorship to students as part of the First Year Advising Team (FYAT), Office of Academic Advising (OAA), cultivating strong leadership and empathy.

PROJECTS

IoT-Based Air Pollution Monitoring & Adaptive Filtration Control

Jan 2025 - Jan 2026

Final Year Design Project

- IoT Technology; Embedded System Design; WiFi-Based Mesh Technology; Multiple Nodal Sensing; Two-Stage Filtration; Air Pollution Control

Self-Powered Biosensor for Real-Time Wireless Health Monitoring

Mar 2026 - Present

Research Seed Grant Initiative (RSGI) by BRAC University Funded Project

- E-Skin; Biomedical Engineering; Human Health Monitoring; Flexible Sensor Technology; Flexible PCB; Self-Powered Biosensor
- Ongoing Research Assistant for this project

EDUCATION

Bachelor of Science in Electrical and Electronic Engineering

Jan 2022 - Jan 2026

BRAC University

- Major: Electronics
- Relevant coursework in Electrical and Electronic Engineering
- 5x Dean’s Lister; 3x VC’s Lister
- CGPA: 3.81

SKILLS

Python	Leadership	Smart Grids
VLSI Design	Embedded System Design	Project Management

ADDITIONAL INFORMATION

- **Languages:** English, Bangla.
- **Secondary School Certificate (SSC):** Rajshahi Board | Square School and College | GPA: 5.00 | 2018
- **Higher Secondary Certificate (HSC):** Rajshahi Board | Square School and College | GPA: 5.00 | 2020

PUBLICATIONS

Towards smarter grids: A systematic review of wide area monitoring enhancing stability, protection, and promoting environmental sustainability

Energy Conversion and Management: X

- Q1 (Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3
- DOI: <https://doi.org/10.1016/j.ecmx.2025.101424>

AI-powered cybersecurity for smart grid communication: A systematic review of intrusion detection and threat mitigation systems

Energy Conversion and Management: X

- Q1 (Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3
- DOI: <https://doi.org/10.1016/j.ecmx.2025.101416>

State-of-charge-aware charging opportunity detection for electric vehicles using data-driven learning and digital twin simulation

Energy Conversion and Management: X

- Q1 (Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3
- DOI: <https://doi.org/10.1016/j.ecmx.2026.101775>

The next generation of energy storage for smart and sustainable power grids: A systematic review

Energy Conversion and Management: X

- Q1 (Top 7%) | Impact Factor: 8.78 | CiteScore: 11.3
- DOI: <https://doi.org/10.1016/j.ecmx.2026.101771>